From Mass Spectrum to Molecule

How the current public model for the Enveda CASMI 2026 Kaggle challenge works, and what our strategy has found so far

Technical brief · October 18, 2026 · competition runs Sep 14 – Dec 14 2026 ($50k, top 5 paid)

In short. The best public pipeline (public score 0.450) is a *retrieval* system: compute the unknown molecule's mass, list every known structure with that mass, score each one with several kinds of spectral evidence, fuse the rankings, and break ties between same-formula isomers with a model that predicts spectra. We (A) reproduced it, (B) attacked the "never-seen molecule" share and learned it is worth at most ~0.01, (C) found two train/serve bugs in it, and (D) are building a fully rule-compliant pipeline of our own.

1 The problem

A tandem mass spectrometer (LC-MS/MS) turns a molecule into a charged ion, weighs it (the **precursor m/z**, mass-to-charge ratio), then breaks it with gas collisions (20, 40 or 60 eV) and weighs the pieces. The resulting **spectrum** is a list of (fragment m/z, intensity) peaks plus the precursor m/z. Each hidden test molecule (natural-product-like) comes with 1–16 spectra (median 3) from a Bruker timsTOF, possibly at several energies and adducts. The task: return up to **25 ranked structure guesses** (SMILES strings).

Score: MRR@25 (mean reciprocal rank) = average over molecules of 1/rank of the first correct guess: rank 1 → 1, rank 2 → 0.5, not in top 25 → 0. "Correct" = same atom connectivity (first InChIKey block; stereochemistry ignored).

Three kinds of test molecule (true shares hidden; community estimates from leaderboard probes):

- **Class 1** (~16–28%): public reference spectra of the structure exist → library look-up works (MRR ~0.93).
- **Class 2** (~20–45%): structure is in a database (PubChem or COCONUT) but was never measured → retrieve it and rank it above same-mass rivals.
- **Class 3** (~39–52%): new structure, not in PubChem → must be *generated*.

Data and rules. 2.54M training spectra of ~275,810 structures from 11 public libraries (largest: enveda-180, 1.15M timsTOF spectra of drug-like compounds; enveda-np-examples: 250 natural products measured like the test). Code runs offline in a Kaggle notebook (≤ 9 h). For prizes, data and pretrained weights must be public, free, commercially usable and reproducible from public code (no NIST).

A noisy scoreboard. The hidden test has ~400 molecules; the public leaderboard sees only 31% (~124). One molecule moving from miss to rank 1 changes the public score by ~0.008 (standard error ~0.016). On Oct 17: top 0.495, best public notebooks 0.44–0.45, 3,486 teams.

KEY TERMS

Adduct	the charged form actually measured, e.g. $[M+H]^+$ (molecule + proton) or $[M+Na]^+$; ten are allowed. Knowing it turns precursor m/z into the neutral mass M.
Fragment	a charged piece of the broken ion; each one gives a peak.
Isomers	different structures with the same formula, hence the same mass: mass alone cannot separate them.
Fingerprint	a bit vector; each bit says whether a given substructure is present. *Tanimoto* = share of bits

2 How the current public model works

The strongest public notebook ("From Spectra to Structures", v51, public 0.450; community "E006" lineage) stacks many community contributions (Fig. 1). It can only output structures that are in its databases, plus a few from a small generator.

1. **Mass → candidates.** M = median neutral mass over the spectra. Every structure within 10 ppm of M (30 ppm fallback) in a ~711k pool (training structures + COCONUT natural products) is a candidate: typically tens to hundreds, mostly isomers (Fig. 2).
2. **Evidence channels (Engine 1).** Each candidate gets ~100 features:
   - *library*: spectral-entropy similarity of the query to the candidate's own reference spectra (solves class 1);
   - *analog propagation*: training spectra that resemble the query (after a mass shift) vote for candidates structurally similar to their molecules (Tanimoto);
   - *fingerprint model* FPNet: a transformer reads peaks + metadata and predicts the unknown's substructure bits; candidates with matching fingerprints score high;
   - *fragmentation*: share of peak intensity explained by breaking 1–2 bonds of the candidate, plus a learned fragment scorer;
   - a small class-3 generator (chemical transformations of the 6 nearest analogs).

   Four LightGBM gradient-boosted models combine the features; then a popularity prior is added: z(score) + 0.15 × log(PubChem substance + literature counts). Top 60 kept.
3. **PubChem channel.** Fingerprint models screen ~105M PubChem structures in the mass window (z + 0.25 × popularity). Picks are inserted at ranks 2, 4, 6, 8, 10, promoted to rank 1 when the fingerprint margin is large (> 150) or the compound is well documented, and skipped when a strong library match (similarity ≥ 0.9) exists.
4. **Engine 2.** Independent candidates (also ChEBI, LIPID MAPS), own channels, two gradient-boosted rankers blended 0.88/0.12; top 40.
5. **Fusion + forward models.** Reciprocal-rank fusion (each list adds 1/(constant + rank)) after merging duplicates. Inside each formula group, ICEBERG/GLACIER predict every candidate's spectrum; the best match to the measured spectrum moves up (fragment score if they give no output). Engine 2's top pick goes to rank 2.

History: 0.34 (mid Sep, 4-channel retrieval + ranker) → 0.40 (engine fusion) → 0.44 (forward-model re-ranking) → 0.45 (tuned PubChem promotion, deeper Engine-2 trees).

	two molecules have in common.
Analog	a training molecule whose spectrum resembles the query; usually a close structural relative.
Forward model	predicts the spectrum of a given structure (the reverse direction of our task).
ppm	parts per million of the mass: 10 ppm of a 400 Da molecule is 0.004 Da.

Weak spots: almost no class-3 hits; many misses are the *wrong isomer*; tweaks were accepted on the noisy 124-molecule board without cross-validation; licence/reproducibility of several bundled weights is uncertain (some CC-BY-NC-SA, one ranker file of unpublished construction).

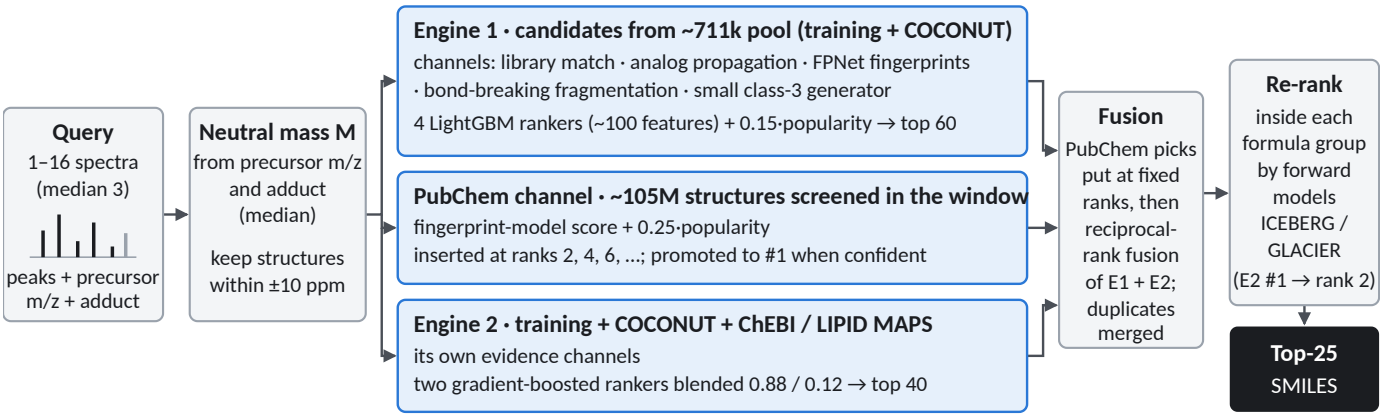


Figure 1. The public model as a pipeline. Blue boxes are the three independent candidate generators/rankers. Everything up to the fusion step chooses *which* known structures enter the list; the forward-model step only reorders candidates that share a formula (isomers), where mass and most evidence channels cannot discriminate.

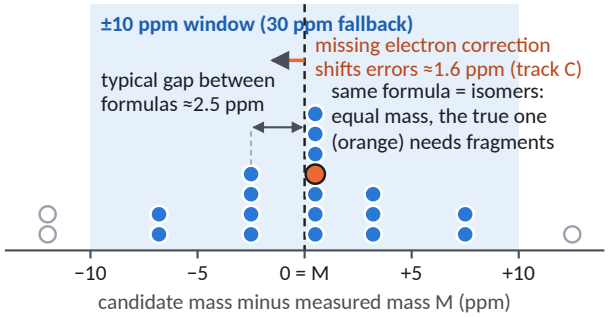


Figure 2. Mass window and candidates (schematic). Each dot is a database structure; columns are molecular formulas. Mass filtering removes most of the database but leaves stacks of isomers, so ranking inside a stack decides the score.

3 Our strategy (first session, Oct 17–18)

A Get on the board. Forked the 0.450 notebook unchanged, keeping its pinned Docker image (teams that ran copies on a newer image lost ~0.04). Score pending.

B Class-3 deep-dive (the largest untapped share).

- **A proxy for "novel".** Checking all 106M PubChem connectivity keys found 16,917 training structures that are *not* in PubChem: real measured molecules, the closest stand-in for class 3. 45% of them split at one amide/ester bond into two PubChem-known pieces (N-acyl amino acids, bile-acid conjugates, fatty-acid esters).
- **NovelEngine**, two generators that output only structures absent from PubChem, COCONUT and training: (i) *analog derivatives*: find similar training spectra by modified cosine (also pairs peaks shifted by the mass difference Δm), then apply transformations (hydroxylation, methylation, glycosylation, chain extension, …) whose mass change equals Δm; from the top-20 analogs this reaches 24% of the proxy; (ii) *conjugates*: join a known acid to a known amine/alcohol (amide/ester/glycoside) whose masses add up, ranked by diagnostic fragments of each half.
- **Site localisation** (our new scoring idea, Fig. 3) raised derivative MRR on the proxy from 0.061 to 0.107; a learned LambdaRank ranker raised proxy MRR from 0.124 to 0.269.
- **Reality check.** A submission of *only* novel candidates scored 0.010 public vs ~0.045 predicted from the proxy: the proxy overstates reach ~4–5×, and class 3 is worth at most ~0.01 as an add-on. A "known-only" run of the same engine scored 0.085.

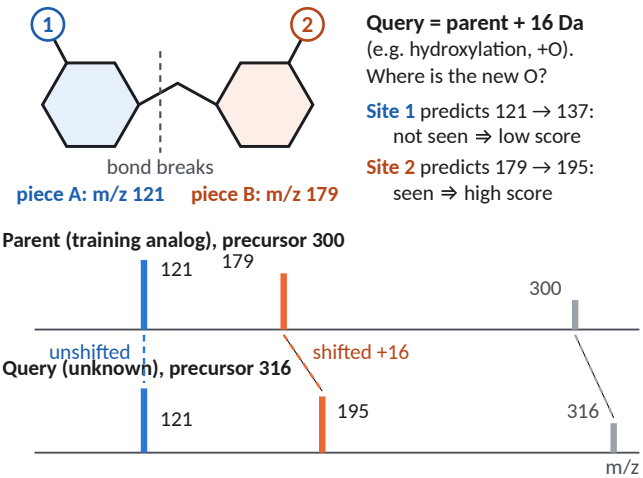


Figure 3. Site localisation (toy numbers). A parent peak that reappears unshifted comes from a piece without the modification; one that

C Bug hunt on the public pipeline (code review by a subagent) found two train/serve mismatches: (1) the ranker feature "number of peaks" is computed on raw test peaks (median 230) but was trained on cleaned peaks (median 52); (2) all models were fitted on enveda-180 masses *without* the electron-mass correction (−0.000548 Da) that the hidden test's positive-mode m/z values carry: a ~1.6 ppm shift in mass-error features at the median mass, against a ~2.5 ppm typical gap between competing formulas (Fig. 2). Two patched variants submitted (scores pending).

D Compliance-first pipeline of our own (prize eligibility: all code and weights ours, reproducible, permitted data).

- **Pool:** 726,198 structures (training + COCONUT Oct-2026 release).
- **Channels:** library entropy similarity, analog propagation, fragment explanation, COCONUT documentation priors (organisms, literature, synonyms), and our own spectrum↔structure model.
- **Own model, a dual encoder** (Fig. 4): it learns to place a spectrum and its true structure close together and up to 64 same-mass "hard negatives" far away (contrastive loss), while also predicting fingerprint bits. After one CPU epoch it ranks held-out structures among their same-mass candidates with **MRR 0.51 (top-1 36%)**; longer CPU and GPU runs are in progress.
- **Ranker:** LambdaRank (a LightGBM model trained to order lists) on simulated queries from 8,000 training structures (class 1: hide only the query spectra; class 2: hide every spectrum of the structure), validated on 1,500 disjoint structures (250 enveda-np-examples, 750 natural products, 500 others). Simulations use the *measured* mass (the true mass would leak the answer) and apply the electron correction to mimic the test.

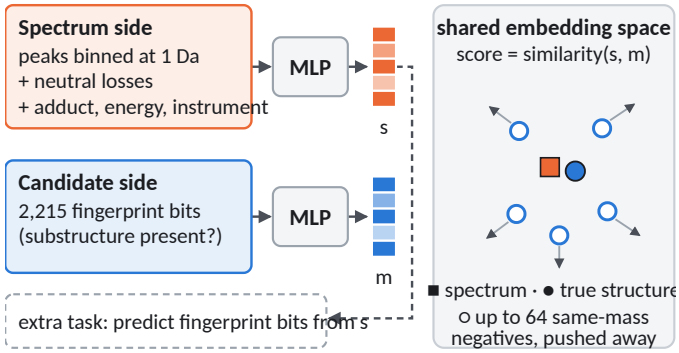


Figure 4. Our dual-encoder model. Two small networks map a spectrum and a candidate structure to vectors s and m. Training pulls the true pair together and pushes same-mass decoys apart, so at test time candidates in the mass window are ranked by similarity to the query.

4 Results so far & lessons

	Experiment	Measured on	Result
A	Fork of 0.450 notebook	public board	*pending*
B	Analog-derivative reach	proxy, top-20 analogs	24%
B	Site localisation	proxy (derivatives)	MRR 0.061 → 0.107
B	Learned ranker	proxy	MRR 0.124 → 0.269
B	Novel-only submission	public board	0.010 (~0.045 predicted)

reappears shifted by Δm comes from a piece that carries it. Each candidate site is scored by how consistent all peaks are with it.

	Experiment	Measured on	Result
B	Known-only, same engine	public board	0.085
C	Two bug-patched variants	public board	*pending*
D	Dual encoder, 1 CPU epoch	held-out, same-mass	MRR 0.51, top-1 36%

- **Validate proxies against the leaderboard early:** "absent from PubChem" training molecules were not like the hidden class 3 (reach overstated ~4–5×).
- **The points are in classes 1–2:** putting the right *isomer* first among known structures.
- **Check train/serve consistency:** a 1.6 ppm offset is large next to a 2.5 ppm formula gap.
- **Select with held-out validation,** not only the 124-molecule board (~0.008 per molecule).

5 Next steps (track E)

Compare the variants on the board; finish our own pipeline (target ≥ 0.45 while fully compliant); add the class-3 engine only where the main list is weak; pick the two final submissions by validation + board. **Why:** if the hosts enforce the rules (they "reserve the right to check all submissions"), strong compliant entries may outrank non-reproducible ones.